Explore Sinusoids

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Use the `starter.m` file to complete the following exercises. Be sure to change the placeholder lines of the starter file to match specific information for this assignment.

Represent continuous-time signals using a dense set of samples (e.g., at least 100 samples per period for a sinusoid) and display them using plot. Display discrete-time signals using stem or comb. Be sure to label all axes and include units (e.g. "Time (msec)").

1. Plot a continuous-time sinusoidal signal $x(t) = 5\cos(2\pi ft + \phi)$ volts on the interval $-1 \leq t \leq 1$ seconds.
a) $f = 1, \phi = 0$
b) $f = 1, \phi = -\pi/4$
c) $f = 2, \phi = -\pi/4$
d) $f = 5, \phi = -\pi/4$
e) $f = 10, \phi = -\pi/4$
Comment on how the frequency f and phase affect the nature of the sinusoid.

2. Graph the samples of a discrete-time sinusoid $x[n] = 3\cos(\omega n + \phi)$ on $-10 \leq n \leq 90$.
a) $\omega = \pi/50, \phi = \pi/4$
b) $\omega = \pi/5, \phi = 0$
c) $\omega = \pi/2, \phi = 0$
d) $\omega = 11\pi/5, \phi = 0$
e) $\omega = 21\pi/5, \phi = 0$
Comment on how the frequency ω affects the nature of the sinusoid. What are the conditions on ω for $x[n]$ to be a unique function of ω?

3. Define a vector of time values t that go from 0 to 1 sec at intervals of 0.001 sec using the colon operator, and compute a complex sinusoid vector c = exp(j*3*pi*t) where j = sqrt(-1). Define another complex sinusoid having twice the frequency of c, that is, s = exp(j*6*pi*t).
a) Use the colon operator to display the values of elements 50-54 of c.
b) Compute the sum of c and s: y = 2*c - s. Plot the real and imaginary parts of y using real and imag.
c) Compute the element-by-element product of c and s using the .* operator to perform element-by-element multiplication (instead of vector inner product). Plot the magnitude of the product using abs. Can you explain why the result looks like it does? Plot the phase of the product using angle.